

Monotone functions on an interval and invertibility



Outline

Monotone functions on an interval

Continuity and invertibility

Monotone functions on an interval



Introduction: monotone functions on an interval

- ▶ In this section we study **monotone functions** on an interval, **not necessarily continuous**
- ▶ We will prove that a monotone function always has **right and left limits** at every point
- ▶ The result is based on the **completeness axiom of $\mathbb{R}$** and extends to functions the monotonicity theorem for sequences
- ▶ As a consequence, the only possible discontinuities of a monotone function are **jump discontinuities**

Monotone functions on an interval

- ▶ We now deal with monotone functions on an interval (and **not necessarily continuous**), and the next theorem (based on the completeness axiom of $\mathbb{R}$) can be seen as an extension to functions of the monotonicity theorem for sequences.

Theorem 1: Monotonicity theorem for functions

Let $f : (a, b) \rightarrow \mathbb{R}$ be a monotone function. Then for every $c \in (a, b)$ the right and left limits as $x \rightarrow c$ exist and are finite; at the two endpoints a, b the right limit (at a) and the left limit (at b) exist, possibly infinite.

Proof (1/6)

Suppose f is increasing in (a, b) and let $c \in (a, b)$, that is, a point inside the interval.

We prove that the following limit exists and is finite:

$$\lim_{x \rightarrow c^-} f(x) = \sup \{f(x) : x \in (a, c)\}.$$

We set

$$\ell = \sup \{f(x) : x \in (a, c)\}$$

Note that ℓ exists and is finite by the supremum property, since $f(c)$ is an upper bound of the set $\{f(x) : x \in (a, c)\}$.

Hence we need to prove that

$$\lim_{x \rightarrow c^-} f(x) = \ell$$

Proof (2/6)

So let $\{x_n\}$ be any sequence in (a, c) such that $x_n \rightarrow c$, and let us prove that $f(x_n) \rightarrow \ell$, i.e., that for every $\varepsilon > 0$ we eventually have

$$\ell - \varepsilon < f(x_n) < \ell + \varepsilon.$$

The second inequality is obvious, since $f(x_n) \leq \ell$ by definition of ℓ , because $x_n \in (a, c)$.

To prove the first one, we observe that, since $\ell - \varepsilon$ is less than ℓ , i.e., less than the least upper bound of $\{f(x) : x \in (a, c)\}$, it is not an upper bound of this set; therefore there exists a point

$$\tilde{x} \in (a, c) \text{ such that } f(\tilde{x}) > \ell - \varepsilon$$

Proof (3/6)

Since f is increasing, it follows that

$$f(x) > \ell - \varepsilon, \quad \forall x \in (\tilde{x}, c)$$

On the other hand, by hypothesis

$$x_n < c, \quad \forall n \quad \text{and} \quad x_n \rightarrow c \quad \text{as} \quad n \rightarrow +\infty$$

therefore $x_n \in (\tilde{x}, c)$ eventually. Hence

$$f(x_n) \geq \ell - \varepsilon \quad (\text{eventually})$$

which is what we needed to prove.

Similarly, one can prove that the following limit exists and is finite:

$$\lim_{x \rightarrow c^+} f(x) = \inf \{ f(x) : x \in (c, b) \}.$$

Proof (4/6)

As for the limits at the two endpoints of the interval, we prove that the following limit exists:

$$\lim_{x \rightarrow b^-} f(x)$$

We set (as before)

$$\ell = \sup \{f(x) : x \in (a, b)\}$$

however, in this case ℓ could also be $+\infty$ (we cannot claim that $f(b)$ is an upper bound of the set, because the function is not defined at b). We therefore proceed by cases.

In the case $\ell < \infty$ the previous proof can be repeated.

Proof (5/6)

If instead $\ell = \infty$, the argument changes. From the hypothesis

$$\sup \{f(x) : x \in (a, b)\} = +\infty$$

it follows that for every $M > 0$ there exists $\tilde{x} \in (a, b)$ such that $f(\tilde{x}) > M$.

By the monotonicity of f , then

$$f(x) > M, \quad \forall x \in (\tilde{x}, b)$$

hence, taking any sequence $x_n \rightarrow b$, we have $x_n \in (\tilde{x}, b)$ eventually, and therefore

$$f(x_n) > M, \quad \text{eventually,}$$

Proof (6/6)

therefore

$$\lim_{n \rightarrow +\infty} f(x_n) = +\infty$$

In the same way one proves that the following limit exists:

$$\lim_{x \rightarrow a^+} f(x)$$



Discontinuities of monotone functions

A consequence of the monotonicity theorem is that if a function is monotone on an interval (a, b) , its possible points of discontinuity in (a, b) are necessarily **jump discontinuities**, except for the endpoints a, b , where there may also be a **vertical asymptote**.

Summary

Monotonicity theorem for functions:

- ▶ If $f : (a, b) \rightarrow \mathbb{R}$ is monotone, at every $c \in (a, b)$ the right and left limits exist and are **finite**
- ▶ At the endpoints a, b the right and left limits exist, possibly infinite
- ▶ For f increasing: $\lim_{x \rightarrow c^-} f(x) = \sup\{f(x) : x \in (a, c)\},$
 $\lim_{x \rightarrow c^+} f(x) = \inf\{f(x) : x \in (c, b)\}$

Consequence:

- ▶ A monotone function on an interval can only have **jump discontinuities** (and vertical asymptotes at the endpoints)

Continuity and invertibility



Introduction: continuity and invertibility

- ▶ In this section we study the link between **monotonicity**, **continuity** and **invertibility**
- ▶ We will see that, in general, an invertible function is not necessarily monotone
- ▶ However, for a function that is **continuous on an interval**, being invertible is equivalent to being **strictly monotone**, and the inverse is again continuous
- ▶ This result completes the proof of the continuity of the **elementary functions** (logarithms, inverse trigonometric functions)

Monotonicity and invertibility

- ▶ We have seen the theorem stating that if a generic function with domain D is strictly monotone, then it is invertible.
- ▶ We also know that the converse is not true in general: there exist functions that are invertible on an interval but not monotone.



Example: invertible but not monotone function (1/2)

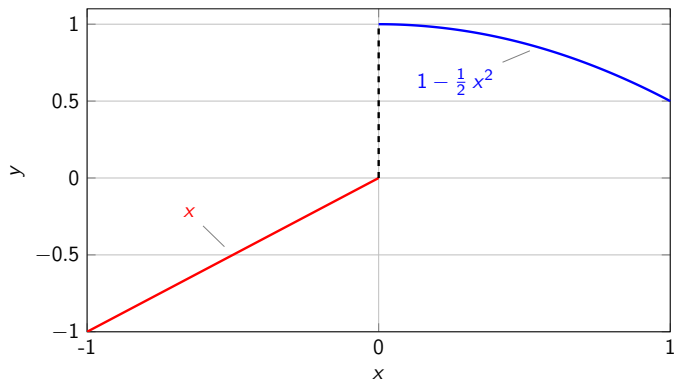
Consider for example

$$f(x) = \begin{cases} 1 - \frac{1}{2}x^2 & \text{if } 0 < x \leq 1, \\ x & \text{if } x \leq 0 \end{cases}$$

This function satisfies the invertibility condition, which requires the graph of f to be intersected at most at one point by every line parallel to the x -axis, but it is **not monotone**.

Its graph is shown in the next slide.

Example: invertible but not monotone function (2/2)



Invertibility theorem

- ▶ If we add the hypothesis of continuity and require the domain to be an interval, being strictly monotone becomes a necessary and sufficient condition for invertibility, as stated by the following theorem.

Theorem 2: Invertibility of monotone and continuous functions

Let $f : I \rightarrow \mathbb{R}$ be a function defined on an interval I .

If the function f is continuous, then it is invertible on the interval I if and only if it is strictly monotone.

In this case its inverse function is also strictly monotone and continuous.

Proof (1/4)

We already know that if f is strictly monotone, it is invertible (regardless of the hypotheses that f is continuous and that it is defined on an interval).

We show that the converse holds, i.e., that if f is continuous and invertible, then it is strictly monotone.

Suppose by contradiction that the function is not strictly monotone; then there exist three points

$$x_1 < x_2 < x_3 \quad (\text{in the interval } I)$$

such that

$$f(x_1) < f(x_2) \quad \text{and} \quad f(x_2) > f(x_3)$$

or such that

$$f(x_1) > f(x_2) \quad \text{and} \quad f(x_2) < f(x_3).$$

Proof (2/4)

Suppose the first of the two alternatives holds (the other case is handled similarly). Let us compare the values $f(x_1)$ and $f(x_3)$; they cannot be equal because f is invertible by hypothesis, hence

$$f(x_1) < f(x_3) \quad \text{or} \quad f(x_1) > f(x_3).$$

Again, suppose the first of the two alternatives holds (the other case is handled similarly). So we know that:

$$x_1 < x_2 < x_3 \quad \text{and} \quad f(x_1) < f(x_3) < f(x_2).$$

Proof (3/4)

Since f is continuous, by the intermediate value theorem there exists

$$x_0 \in (x_1, x_2) \text{ such that } f(x_0) = f(x_3).$$

Since $x_0 \neq x_3$ (because $x_1 < x_2 < x_3$), it follows that f cannot be invertible, a **contradiction**. This proves the first part of the theorem.

Proof (4/4)

Now let f be a continuous, strictly monotone and hence invertible function on I , and let g be its inverse function, which is also strictly monotone and invertible. We prove that g is continuous.

- ▶ By what we observed after the monotonicity theorem, the strictly monotone function g is either continuous or has jump discontinuity points.
- ▶ In the latter case the image of g is not an interval (but the union of at least two disjoint intervals), which is a **contradiction** because this image is I . Hence g is continuous.



Remarks on the proof

- ▶ Note that in the proof of the previous theorem we used both the intermediate value theorem and the monotonicity theorem for functions. Moreover, we implicitly used the completeness axiom of $\mathbb{R}$.
- ▶ The theorem just proved means, in particular, that:

a continuous and invertible function on an interval has a continuous inverse function.

Continuity of the elementary functions

- ▶ This fact completes the proof of the continuity theorem for elementary functions:
 1. the continuity of the function a^x implies the continuity of the function $\log_a x$;
 2. the continuity of the functions $\sin x$, $\cos x$ and $\tan x$ implies the continuity of the functions $\arcsin x$, $\arccos x$ and $\arctan x$.

Summary

Monotonicity and invertibility:

- ▶ Strictly monotone $\implies$ invertible, but in general the converse does not hold

Invertibility theorem:

- ▶ If $f : I \rightarrow \mathbb{R}$ is continuous on an interval I : f invertible $\iff f$ strictly monotone
- ▶ The inverse is again strictly monotone and **continuous**
- ▶ Proof: intermediate value theorem and monotonicity theorem

Consequence:

- ▶ Continuity of $\log_a x$, $\arcsin x$, $\arccos x$, $\arctan x$

Final summary

- ▶ A **monotone** function on an interval always has right and left limits: finite at interior points, possibly infinite at the endpoints
- ▶ Its discontinuities can only be **jump discontinuities**
- ▶ For a function **continuous on an interval**: invertible $\iff$ strictly monotone, and the inverse is continuous
- ▶ This yields the continuity of **logarithms** and of the **inverse trigonometric functions**